Design Model: Stepped Transmission Shaft Ø60×200mm

APPROVED: FULLY COMPLIANT

Part Type: SHAFT | Material: Stainless Steel 304 (AISI 304 / 1.4301)

Validation Date: 10/14/2026, 6:22:26 AM | Lead Engineer: Viraj Bansod

Factor of Safety: 119.74x (Min Required: 1.5x)

1. Geometric Integrity & Constraint Checks (1 Rules Evaluated)

Rule / Metric	Severity	Validation Feedback & Suggested Fix
Valid Parametric Geometry	PASS	All geometric relationships, clearance envelopes, and ma

2. Structural Stress & Factor of Safety Analysis

Shaft Torsional, Bending & Deflection Analysis — **FoS: 119.74x [SAFE]**

At 5 kW @ 1440 RPM, the shaft transmits 33.16 N·m. Operating at a von Mises stress of 1.8 MPa against a materi

Engineering Note: Design is structurally sound for continuous power transmission (FoS = 119.74).

ISO 281 Rolling Bearing L•€ Rating Life — **FoS: 59.30x [SAFE]**

Under 1200 N radial load at 1440 RPM, the Ø60 mm bearing has an estimated ISO 281 basic rating life of 2,408,5

Engineering Note: Bearing life exceeds industrial 20,000h machine requirement.

3. Design For Manufacturing (DFM) & GD&T Inspection

Process: CNC Lathe Turning (Single-Setup with Live Tooling)

Estimated Machining Time: 18 minutes

Tooling Accessibility: Good (Score: 92/100)

Min Wall Thickness: 6 mm | Standard: ISO 2768

Key GD&T Tolerances (ISO 1101 / ASME Y14.5)

[CYLINDRICITY] Main Bearing Journal (Ø30 mm)	Tolerance: ±0.008 mm	Datum: A	Maintains uniform hydrod
[RUNOUT] Shaft Centerline Axis Runout	Tolerance: ±0.015 mm	Datum: A-B	Prevents dynamic imbalan
[POSITION] Keyway Centerline Symmetry	Tolerance: ±0.03 mm	Datum: A-B	Ensures parallel alignme

Manufacturing Cost & Mass Summary:

Total Mass: 4.56 kg | Material: Stainless Steel 304 (AISI 304 / 1.4301)

Estimated Production Cost: INR 2,259

FORMAL ENGINEERING VALIDATION CERTIFICATION

Validated against ISO 2768 (General Tolerances), ISO 2048 (Shaft Keys), and ASME Boiler & Pressure Vessel codes.

Lead Designer: Viraj Bansod | CADVANTA Automated Engineering Suite

`[VERIFIED STAMP: DOC-ENG-MV7Q38J3]`